

Transporter

Transporter

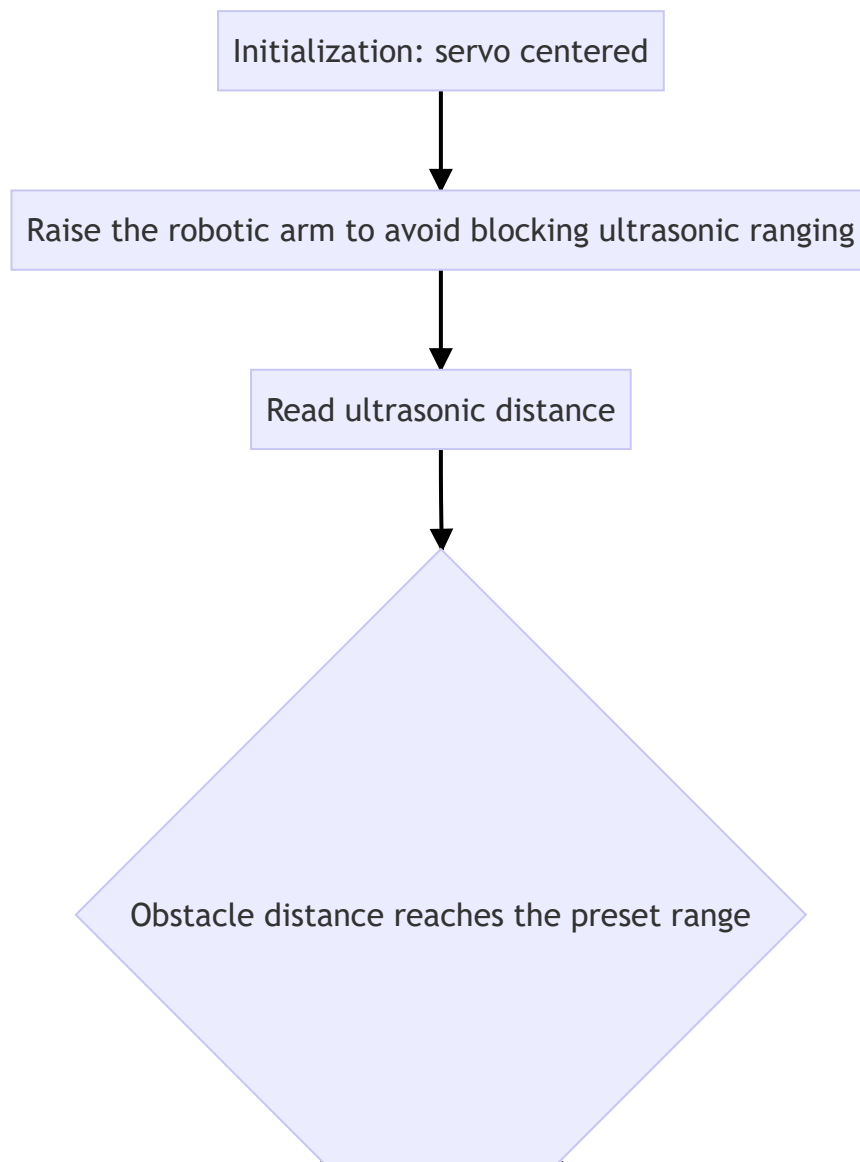
1. Experiment Purpose
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4. Experiment Steps
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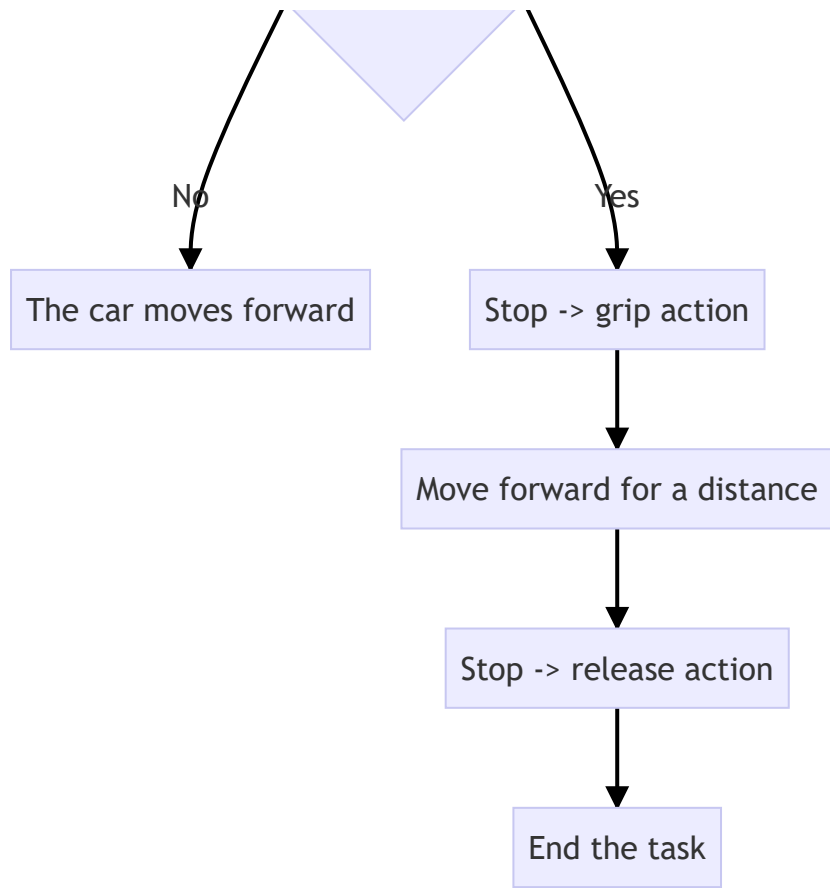
1. Experiment Purpose

In this section, we will use the ultrasonic sensor together with servo control to achieve the "transporter" effect. This section provides two example programs: "14.Fixed_Distance_Transport.hex" and "14.Transporter.hex". Motor control has been added to "14.Transporter.hex".

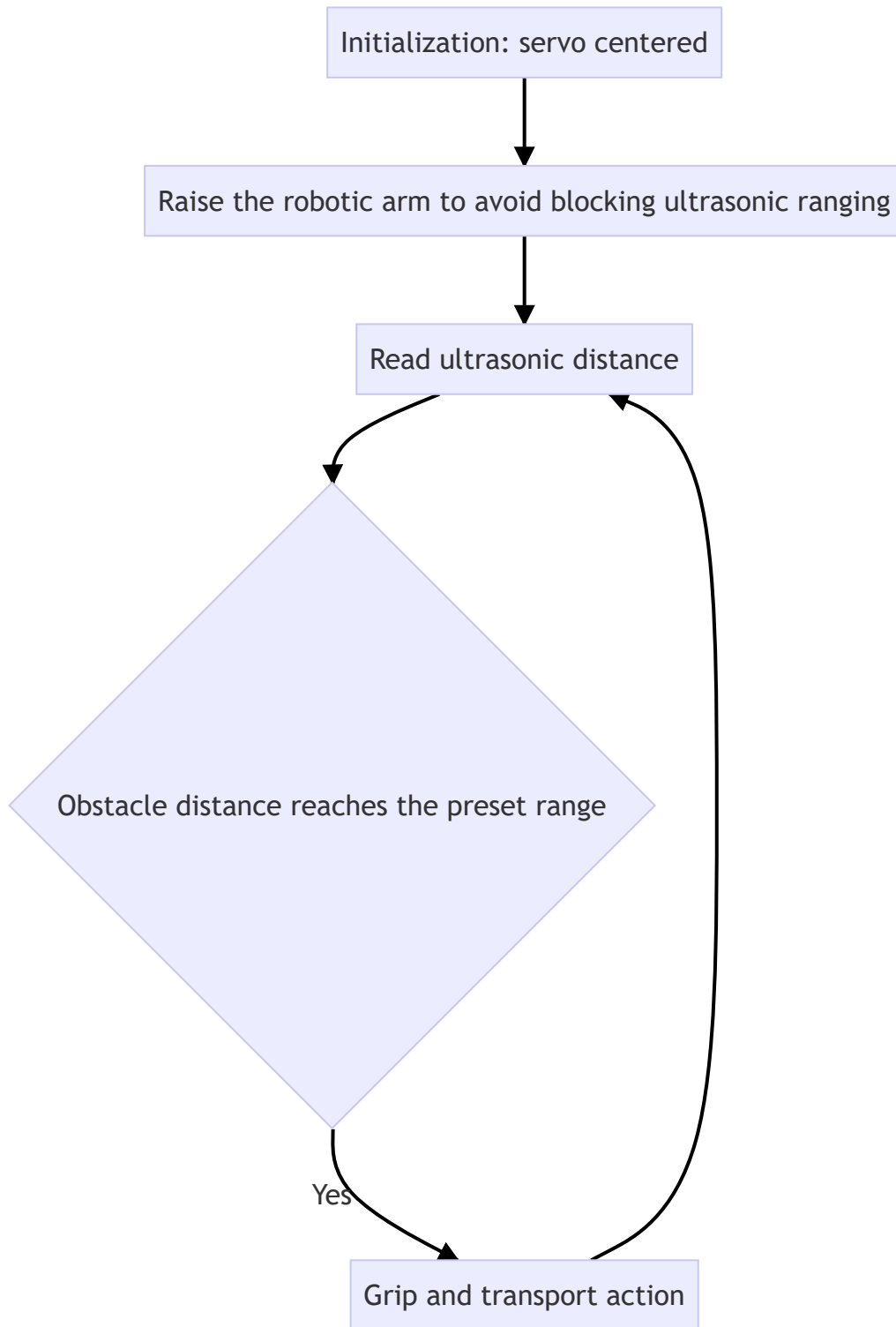
2. Implementation Principle

Transporter

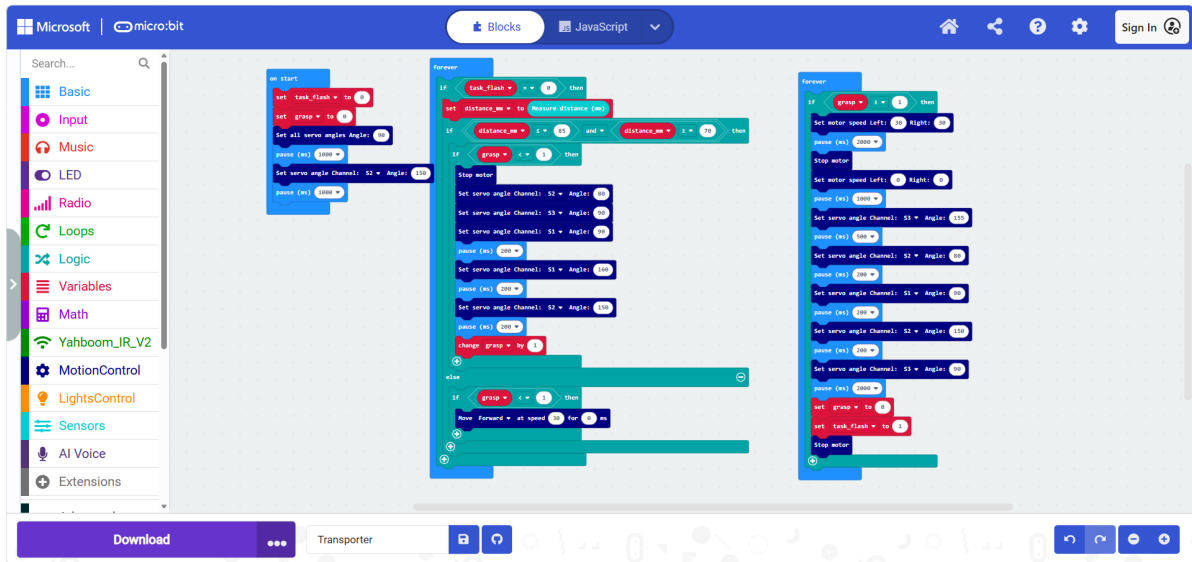




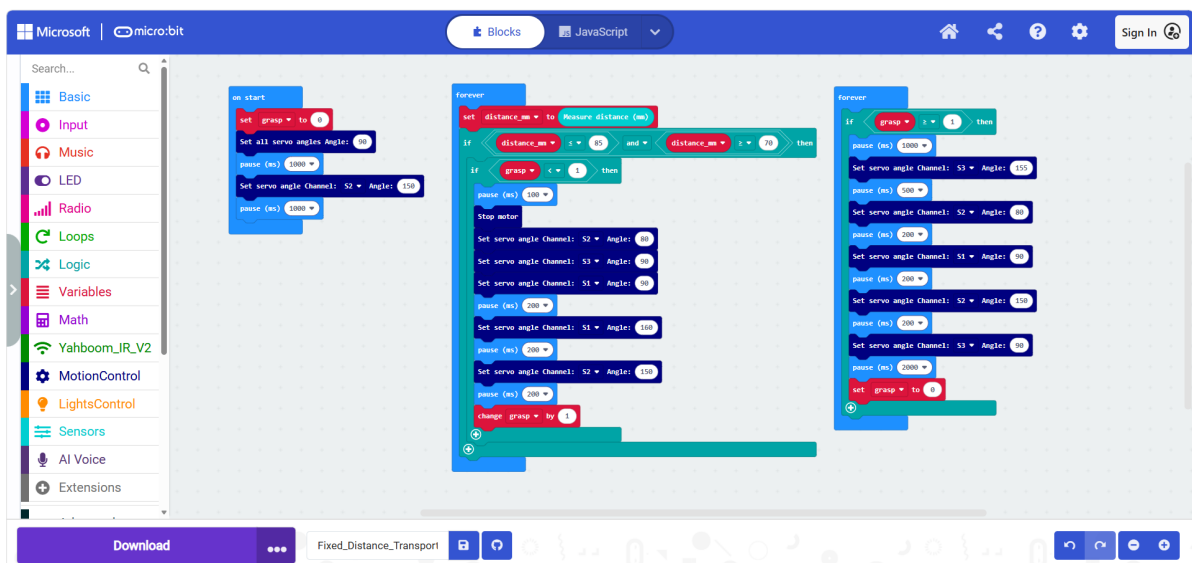
Fixed-Distance Transport



Transporter program is as follows.



Fixed-Distance Transport program is as follows.



The positions of the blocks have already been introduced in previous tutorials, so we will not go into too much detail here.

4. Experiment Steps

1. Before flashing, turn off the car switch (switch it to OFF), and connect the Micro USB cable to the Micro:bit. Note: not the Charging interface on the expansion board.
2. Flash `14.Transporter.hex` or `14.Fixed_Distance_Transport.hex` to the Micro:bit. For the flashing method, refer to section "4. Flashing Programs" in "2. Introduction to Graphic Programming"
3. Turn on the car switch (switch it to ON).

5. Experiment Phenomenon

Due to mechanical errors in the motors and tire slippage, the car cannot drive perfectly straight. If the block is not placed directly in front of the ultrasonic sensor, the gripping effect will be affected. In this case, you need to manually place the block directly in front of the car.

Fixed-Distance Transport: Place the cube EVA block directly in front of the ultrasonic sensor (within the gripping range of the robotic arm), and the car will complete the grip-and-transport action. **The car's motors do not move in this mode.**

Transporter: The car will drive forward. Place the cube EVA block directly in front of the ultrasonic sensor; when the distance between the car and the block is within the set gripping range, the car will complete the gripping action. (The preset gripping distance can be adjusted according to the actual situation.)

After gripping the block, the car lifts the block and moves forward for a distance.

Finally, the car stops and places the block at the front-left, ending the program.